

Problem: can a complex model time the market?

Task. Predict next month's excess return of the CRSP value-weighted index from the 15 Goyal–Welch predictors G_t (dp, dy, ep, de, svar, b/m, ntis, tbl, lty, ltr, tms, dfy, dfr, infl, lagged return), 1926–2020.

Consensus (Goyal & Welch, 2008): out of sample these predictors do not beat the historical mean $\Rightarrow$ use simple models.

Puzzle. ML models have $P \gg T$. Textbook statistics: overfitting. KMZ: complexity is a *virtue*.

Data (voc_data.py, Sec. V.A):

- ▶ returns / trailing 12-month uncentered std
- ▶ predictors / expanding std, ≥ 36 months
- ▶ only past information $\Rightarrow$ sample starts in 1930

Model (eqs. 1–3): unknown f approximated by P nonlinear signals

$$R_{t+1} = f(G_t) + \varepsilon_{t+1} \approx \sum_{i=1}^P S_{i,t} \beta_i + \varepsilon_{t+1}$$

Timing strategy (eq. 6): position = forecast

$$\pi_t = S_t' \hat{\beta}, \quad R_{t+1}^\pi = \pi_t R_{t+1}$$

Evaluation (eq. 5): economic value, not only accuracy

$$SR = \frac{E[R_{t+1}^\pi]}{\sqrt{E[(R_{t+1}^\pi)^2]}}$$

Problem: why a bad R^2 can still pay

Prop. 1 (infinite sample, true β known):

$$SR \rightarrow \frac{1}{\sqrt{3 + (b_* \psi_{*,1})^{-1}}} < \sqrt{1/3} \approx 0.58$$

Even the infeasible strategy has a modest ceiling.

MSE decomposition (eq. 8):

$$MSE(\hat{\beta}) = E[R_{t+1}^2] - 2 \underbrace{E[\hat{\pi}_t R_{t+1}]}_{\text{timing exp. return}} + \underbrace{E[\hat{\pi}_t^2]}_{\text{timing leverage}}$$

Large, noisy positions ruin MSE and R^2 , yet the expected return of the strategy can stay positive.
 $\Rightarrow$ negative $R^2 \neq$ no economic value.

The claim (Fig. 7, Table I):

- ▶ $P = 12,000$ random Fourier features
- ▶ $T = 12$ months of training data,
 $c = P/T = 1,000$
- ▶ ridge $z = 10^3$
- ▶ out-of-sample $SR = 0.47$, 1930–2020
- ▶ more complexity $\Rightarrow$ higher return and Sharpe

Our data check: same setting, 5 seeds
 $\Rightarrow SR \approx 0.49$.

Critique 1–2: momentum and a simple benchmark

Kernel view (Nagel, 2025): with $P \gg T$ the forecast is a weighted average of the last 12 returns

$$\hat{R}_{t+1} \approx \sum_{s=t-12}^{t-1} w(G_t, G_s) R_{s+1}$$

Persistent $G \Rightarrow$ recent months most similar $\Rightarrow$ volatility-timed time-series momentum.

- ▶ Nagel: vol-timed momentum IR 0.39 vs RFF 0.26; on simulated reversals RFF still bets on momentum and loses
- ▶ Elmore & Strauss (2025): forecasts 97.5% correlated with past 12-month return
- ▶ Cartea & Shi (2025): flip the sign of returns $\Rightarrow$ the “virtue” reverses
- ▶ KMZ, p. 499: acknowledge the link, report alpha over TSMOM

The paper's own Table I ($z = 10^3$, Sharpe):

T	Linear, $P = 15$	RFF, $P = 12,000$
12	0.46	0.47
60	0.44	0.42
120	0.49	0.41

Shrinkage, not 12,000 features, does most of the work.

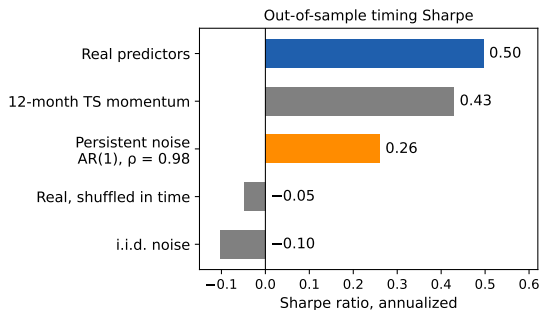
Buncic (2025, rev. 2026): implementation choices handicap simple models; standard linear models reach Sharpe ratios up to 40% higher.

In fairness: KMZ report IR 0.26 ($t = 2.5$) of RFF over the linear model at $T = 12$.

Critique 3: how could this be overfit?

1. $\gamma = 2$ is a choice. Paper: “insensitive”. Our rerun:
 $\gamma = 0.5, 1, 2, 4 \Rightarrow SR = 0.27, 0.36, \mathbf{0.49}, 0.35$
2. $T = 12$ window. Short window + persistent predictors = momentum by construction; RFF Sharpe falls to 0.42 / 0.41 at $T = 60$ / 120
3. **Standardization.** 12-month uncentered vol for returns, expanding std for predictors, 36-month burn-in: each step is a free choice
4. **Shrinkage grid.** Headline $z = 10^3$ is the top of $\log_{10} z \in [-3, 3]$; no out-of-sample rule for choosing z
5. **One sample.** US, 1930–2020; returns halve after 1975 (p. 499). Our rerun: SR 0.57 before, 0.42 after
6. **Data timing.** Inflation for month t is published mid- $(t + 1)$ but used at t (fn. 33; results hold without it, IA)
7. **Seeds.** Averaging 1,000 RFF draws hides single-draw variance (ours: 0.46–0.51 per seed, small)

Our check: noise predictors & verdict



RFF, $P = 12,000$, $z = 10^3$, $T = 12$, $\gamma = 2$, 1930–2020;
noise: mean of 4–6 draws. Code: `critique_checks.py`.

Result. Slow-moving pure noise earns $SR \approx 0.26$, about half of the real result; i.i.d. or time-shuffled inputs earn ≤ 0 .

$\Rightarrow$ persistence + 12-month window create momentum mechanically.

But real predictors keep alpha over 12-month TSMOM (IR 0.33, $t = 3.1$).

Holds up: negative $R^2 \neq$ no economic value; shrinkage is essential near $P \approx T$.

Contested: that complexity itself creates predictability, and that the signal comes from the predictors rather than from momentum.

Reply: Kelly & Malamud (2025), “Understanding the Virtue of Complexity”.